

Can AI Make Better Drugs? Not on Wall Street's Timeline

The technology has arrived in the drug lab. But should investors pay up for it?

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Genentech headquarters in South San Francisco, California.

Artificial intelligence has proven it can speed up what humans already know how to do: write software, summarize earnings calls and automate routine tasks.

Whether it can help discover what humans don't yet know is the trillion-dollar question. Nothing would validate AI's promise more than improving the process of creating medicines for diseases like Alzheimer's and cancer.

Yet drug development is also where AI collides with the stubborn reality of human biology. Wall Street is waiting for evidence that AI can change the economics of making drugs. Until that happens, investors have little reason to reward drugmakers the way they have rewarded the companies building AI tools.

In the lab, AI is already delivering a tangible leap in efficiency, predicting how proteins fold, identifying potential drug targets, and screening millions of molecules virtually. Traditionally, scientists have selected drug targets largely by hand, drawing on the body of research they know best to decide which proteins might matter in a disease, then spending months or years testing those hypotheses in the lab.

At Roche's Genentech, computational biologist Aviv Regev has built what she calls a "lab in the loop." AI models predict promising targets and molecules, researchers test those predictions experimentally, and the resulting data are fed back into the models to improve their next round of predictions.

Regev says the approach has expanded the range of research programs scientists can realistically pursue. Its advantage isn't that AI reasons better than humans, she says, but rather that it can absorb and draw on vastly more biological knowledge than any individual researcher. "AI is not smarter," she says in an interview. "But what helps our scientists is that it encodes information very, very broadly."

There is no doubt astonishing science under way. The question is whether any of it reaches the bottom line. In recent decades, technologies such as genomics, automation and high-throughput screening expanded scientists' ability to generate and test ideas, yet never reliably lifted the return on each research dollar.

The problem isn't a shortage of ideas. It is turning those ideas into medicines that work in people. Only about one in 10 drug candidates that enter human trials ultimately reaches the market, with many failures coming after years of research and billions of dollars of investment.

In computing, Moore's Law captures a dynamic of innovation driving exponential gains in chip performance. But drug research has in a sense moved in the opposite direction: slower and more expensive over time. Scientists gave that reversal a name: Eroom's Law, Moore's Law spelled backward.

Jack Scannell, who co-wrote the 2012 paper that coined the term, argues that part of the problem is simple diminishing returns: Each new drug has to outperform the ones that came before it, including cheap generic medicines that already work. In theory, this is the kind of wall AI was built to break, by searching through the accumulated knowledge of science and uncovering possibilities no human researcher would have found.

But finding possibilities isn't the same as proving they work. Cell cultures, lab animals and computer models remain imperfect stand-ins for the human body. Scannell puts the challenge vividly: Training an AI on today's data can be "like trying to train your Waymo for San Francisco by getting a frog to ride a bike around Albuquerque." Autonomous vehicles work because real cars have logged millions of real miles. Medicine has never had anything close to a clean map of the human body.

That is why even AI's believers acknowledge that proving its potential will take several more years. Eric Kauderer-Abrams, who leads life sciences at Anthropic, says that AI will bend the curve by attacking multiple bottlenecks at once to boost a drug's clinical probability of success. Still, speaking just as Anthropic launched the new platform Claude Science, he notes the industry is only in the "second inning."

For investors, that creates a dilemma. It is hard to give drugmakers credit when the cost of bringing a medicine to market has done little but climb. Yet AI's advance into drug discovery is too significant to ignore. Pharma CEOs including Eli Lilly's LLY -2.33% decrease; down pointing triangle David Ricks and Novartis's NOVN -0.02% decrease; down pointing triangle Vas Narasimhan are

committing billions to computing power, partnerships and new research platforms. Goldman Sachs estimates that the present value of AI's benefits to drug development could reach as much as \$400 billion over the next decade by shortening development timelines, lowering costs and improving the odds that medicines succeed.

The distinction is between today's economics and tomorrow's possibilities. For now, AI is making research labs more productive without changing the metric that matters most: how many successful drugs emerge from each research dollar. Until that number improves, Wall Street has little reason to rerate the sector.

Over time, that could change. AI might be part of a broader biotech renaissance, but only if the industry can improve the feedback loop between discovery and evidence. That means richer human data, better ways to monitor patients, and clinical trials that can generate answers faster.

Biotech investor Rod Wong, managing partner at RTW Investments, argues that intensifying competition from China could become a catalyst for that shift. China's advantage, he says, is in the speed at which companies can move from research ideas to clinical evidence. That pressure could force the U.S. to rethink a trial system that has become increasingly slow and expensive.

If those pieces come together, Eroom's Law might finally begin to bend. If it does, the winners might not be the companies that build the AI models themselves, but rather those that combine powerful tools with the deepest biological data and the ability to run global drug-development programs. Those advantages favor the largest established pharmaceutical companies, notes Citi healthcare strategist Traver Davis.

But biology runs on its own clock, not the semiconductor cycle. The revolution might yet be real, but we won't know for certain for several more years. In drug development, real and soon are rarely the same thing.